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Capturing Inter-Individual Differences in Cognitive Processes via Hierarchical Clustering in Dynamic Bayesian Networks

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Abstract

People systematically differ in how they integrate prior expectations and new evidence when forming beliefs about others. Yet they are not all different: individuals plausibly fall into groups that share similar cognitive patterns. Bayesian networks are a popular tool for modelling psychological processes, but a single network learned on the population level smooths away inter-individual differences, whereas fitting one network per individual yields noisy estimates that do not generalise. Clustering individuals into groups that share common patterns in belief revision, modelled by similar conditional probability tables, offers a principled compromise. However, choosing an appropriate distance metric to quantify similarity between individuals and a stopping criterion for the clustering process is non-trivial. In this paper, we transfer hierarchical agglomerative lifted model construction to the clustering of individuals in (dynamic) Bayesian networks. On simulated belief updating data, we study how central design choices of such studies affect the results, finding that the number of trials per individual matters substantially more than the number of participants. Overall, we demonstrate that hierarchical clustering recovers the underlying groups reliably while highlighting challenges that arise when transferring such mathematically developed clustering methods to the modelling of cognitive processes.

1. Introduction

Understanding others is a fundamental aspect of social cognition, as individuals continuously interpret incoming social information to form and refine representations of the people around them. Our perception and the interpretation of our environment are shaped not only by bottom-up sensory input but also by top-down processes such as prior expectations and assumptions (Kok et al., 2013; McGovern & Otten, 2024). Mental representations of others are formed to better predict others' behaviour and can change over time as individuals encounter new social information (Mende-Siedlecki et al., 2013). This process, often referred to as belief updating, involves revising prior evaluations in light of new and potentially inconsistent evidence. Depending on the kind of prior information and evidence, the duration of this process can vary (Ferguson et al., 2019; Kammrath et al., 2007) and individuals differ in the integration process which leads to differences in the extent to which they maintain or revise beliefs about others in response to new evidence (Cone et al., 2017).

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People also form internal models of others' cooperativeness during interactions, taking into account not only the other person's personal and behavioural cues but also the situational context (Rossetti et al., 2022). Cooperative person models can cause biased interpretations: ambiguous information can be interpreted in line with expectations and beliefs are maintained, or contradictory evidence may lead to biased updating of beliefs.

Research on inter-individual differences in belief updating is particularly important, as people systematically differ in how they integrate prior social expectations and new evidence, which may explain differences in social sensitivity, bias and adaptability. Grasping these differences is important not only for better understanding perceptual and inferential processes in social cognition but also for improving adaptive human-machine interaction. Despite inter-individual differences, individuals may form distinct groups characterised by similar patterns of belief updating and underlying mental processes, while differing more substantially from other groups or individuals. Identifying such groups could help capture meaningful individual differences while maintaining a level of generalisability across individuals who share similar cognitive processes.

There are various ways to model cognitive processes such as belief updating. Bayesian networks (BNs) provide a principled framework to model probabilistic dependencies between variables (Pearl, 1988), e.g., in domains where many individuals share the same underlying causal structure, ranging from medical diagnosis and risk assessment to cognitive and behavioural modelling. When the parameters of a BN are estimated from data, however, the analyst faces a fundamental tension: a single population-level BN smooths away inter-individual differences, whereas fitting one BN per individual yields noisy estimates that are hard to generalise. Clustering individuals into groups and then learning one BN per group is a principled compromise, yet, despite the central role of BNs in applied probabilistic modelling, the question of how best to perform such clustering when the graph structure is already known has received surprisingly little systematic attention.

Clustering of similar objects in probabilistic graphical models (such as BNs) has been studied from two largely separate angles, and our work sits at their intersection. A first line of work originates from *lifted probabilistic inference*, where indiscernibility in a single model is exploited to compress the model and speed up probabilistic inference therein (Poole, 2003; De Salvo Braz et al., 2005; Niepert & Van den Broeck, 2014). To construct such lifted representations, the Advanced Colour Passing (ACP) algorithm and its approximate variant ε-ACP form the state of the art (Luttermann et al., 2024, 2025). More recently, Speller et al. (2025) build on ε-ACP to introduce Hierarchical Advanced Colour Passing (HACP) by computing ε-values in an agglomerative fashion, thereby producing a sequence of progressively coarser lifted representations. A second line of work directly targets the population-level question of clustering individuals whose data are generated from probabilistic graphical models, for example by fitting mixtures of BNs (Thiesson et al., 1998). Heller & Ghahramani (2005) propose Bayesian hierarchical clustering, which combines agglomerative clustering with a probabilistic model-comparison criterion for deciding when to merge clusters. Such an approach, however, relies on generative assumptions and neglects prior knowledge that is present in our setting, namely that the graph structure of each individual's model is known and identical across individuals, thus reducing the clustering to the conditional probability table (CPT) space. In general, clustering is most often performed by generic algorithms such as k-means (MacQueen, 1967) and DBSCAN (Ester et al., 1996).

Despite the growing body of well-established probabilistic models and clustering approaches, a gap remains between their mathematical development and their application to psychological and cognitive processes. Many models are developed and evaluated primarily at a theoretical or mathematical level, whereas their application to psychological data may involve additional challenges and considerations that are not necessarily addressed during model development. In particular, directly applying such models to actual data — where the underlying causal structure is not known — may not ensure that their assumptions and properties are appropriate for capturing inter-individual differences, groups, or interaction dynamics or may even lead to incorrect conclusions. This study therefore addresses this gap by systematically examining the applicability of probabilistic modelling and concrete clustering techniques to the modelling of cognitive processes during a dyadic interaction, with a particular focus on identifying potential challenges that arise when transferring mathematically developed models to psychological data and on assessing their ability to capture inter-individual differences and identify groups of individuals with similar cognitive patterns.

At the methodological level, we study the clustering of individuals in (dynamic) BNs with identical known graph structures. We present a hierarchical agglomerative clustering approach to cluster individuals directly on their CPTs, while allowing practitioners to define similarity through an application-specific distance metric. We further investigate suitable distance metrics for comparing CPTs and methods for selecting a cut-off threshold when the number of groups is unknown. As a concrete real-world application, we consider a simulated belief updating study inspired by a currently planned cooperative gaze-following experiment with underlying CPTs differing between groups of individuals and with groups varying from clearly separable to nearly indistinguishable.

Specifically, our contributions are as follows. *First*, we transfer hierarchical lifted model construction to hierarchical agglomerative clustering of individuals in (dynamic) BNs with known and identical graph structure, thus reducing the clustering directly to the CPT space in order to efficiently identify groups from participant-specific models obtained in an experiment and to provide a fully inspectable merge sequence. *Second*, we systematically compare different distance metrics for clustering CPTs of individuals considering the specific context of application to cognitive processes. *Third*, we use a simulated but realistic belief updating study oriented towards a real-world application to investigate several strategies for choosing the cut-off threshold ε when the number of groups is not known in advance and benchmark hierarchical clustering against the generic algorithms k-means and DBSCAN. *Finally*, we empirically show that the number of trials per individual has a substantially stronger impact on cluster quality than the number of participants, which has important implications for experimental design. Moreover, we show that hierarchical clustering reliably recovers the underlying groups, although we also point out potential challenges — such as selecting an appropriate cut-off threshold — when applying such modelling techniques to real data.

2. Background

BNs are well-established models to handle uncertainty in various applications, e.g., to model different updating processes such as belief updating. A BN compactly represents a joint probability distribution over a set of variables (Pearl, 1988). A dynamic Bayesian network (DBN) extends this representation to time-evolving processes (Murphy, 2002).

Definition 1 (Dynamic Bayesian Network)**.** A *DBN* is a pair $\mathcal{D} = (\mathcal{B}_0, \mathcal{B}_{\to})$ representing a joint probability distribution over a sequence of random variables $\mathbf{X}_{0:T} = (\mathbf{X}_0, \mathbf{X}_1, \ldots, \mathbf{X}_T)$, where each $\mathbf{X}_t = (X_t^{(1)}, \ldots, X_t^{(n)})$ denotes the state of n random variables at time slice t. Both components $\mathcal{B}_0 = (G_0, \Theta_0)$ and $\mathcal{B}_{\to} = (G_{\to}, \Theta_{\to})$ are *BNs*, that is, pairs consisting of a directed acyclic graph $G = (V, E)$ whose vertices V are random variables and whose edges E encode direct probabilistic dependencies between these random variables, together with a set of conditional probability distributions $\Theta = \{\theta_{X|\mathrm{Pa}(X)}\}_{X \in V}$, where $\theta_{X|\mathrm{Pa}(X)}$ is a conditional probability distribution for variable X given its parents $\mathrm{Pa}(X)$ in G. The prior BN $\mathcal{B}_0$ specifies the distribution $P(\mathbf{X}_0)$ over the initial slice, and the transition BN $\mathcal{B}_{\to}$ is defined over the variables $\mathbf{X}_{t-1} \cup \mathbf{X}_t$, where only the nodes $\mathbf{X}_t$ carry conditional probability distributions, thereby specifying the transition model

$$P(\mathbf{X}_t \mid \mathbf{X}_{t-1}) = \prod_{i=1}^{n} P(X_t^{(i)} \mid \mathrm{Pa}(X_t^{(i)})),$$

where $\mathrm{Pa}(X_t^{(i)}) \subseteq \mathbf{X}_{t-1} \cup \mathbf{X}_t$. Assuming the first-order Markov property and stationarity of $\mathcal{B}_{\to}$, the joint distribution over $\mathbf{X}_{0:T}$ factorises as $P(\mathbf{X}_{0:T}) = P(\mathbf{X}_0) \prod_{t=1}^{T} P(\mathbf{X}_t \mid \mathbf{X}_{t-1})$.

The graph of an example DBN is shown in Fig. 1, where each node represents a variable, the solid arrows define the within-slice edges of a single time slice, and the dashed arrows provide the inter-slice dependencies between consecutive time slices. In the upcoming section, we explain the study setup including the meaning of the variables in more detail.

3. Simulated Data Set About Belief Updating

For simulating data, we stick closely to a currently planned pilot study examining belief updating in a cooperative context based on known cognitive processes. In this experiment, two participants sit opposite each other and share a visual space containing objects of different values (low, medium, high). The initiator looks at one object at a time, while the observer infers the target from the initiator's gaze and rates their confidence (1-4). The initiator then provides feedback (correct vs. incorrect). Correct guesses earn both participants points; incorrect guesses earn none. The initiator is instructed to avoid fixed selection patterns and certain choices of objects are prohibited, such as looking at the same object several times directly in succession. The initiator and the observer are also not allowed to consult one another. However, the initiator is permitted to preferentially look at high-value objects, reflecting cooperative behaviour — which also implies that selecting objects with a lower value is associated with less cooperative behaviour. To systematically manipulate cooperativeness, a confederate replaces the second participant (initiator) and follows predefined object selection patterns.

3.1 Causal Probabilistic Relationships

All associations between the variables are assumed to be causal and are modelled as conditional probabilities. Each variable depends on its parents only (incoming arrows pointing to the variables in Fig. 1). We assume that the observer has a *belief* about the cooperativeness of the initiator (i.e., selecting a high-value object when being cooperative). Depending on this belief and an uncertain *sensory evidence* (the initiator's gaze), the observer makes a *choice* and rates the *confidence* of this

choice. Please note that if belief and sensory evidence are not consistent, confidence will probably decrease and the choice might either be based on the prior belief or the sensory evidence (ambiguous gaze). The congruence between choice and evidence defines the *feedback*. The belief for the next trial ($t + 1$) is updated depending on feedback, choice, and previous belief of the current trial (t). Updating happens for example if the feedback implies that the last choice has been incorrect or if the observer has trusted the sensory evidence (e.g., low-value object) which has not been in line with the belief (e.g., high-value object) and has correctly chosen accordingly (here low-value object). If that choice is correct, the observer might update the belief for the next trial. Conversely, the previous belief might be reinforced if that choice had turned out to be wrong (i.e., feedback incorrect).

In our setting, each time slice models one trial of the experiment, the within-slice CPTs encode the dependencies between all variables within one trial t, and the inter-slice CPTs encode how the prior belief is updated in the next trial $t+1$ based on trial t. Individual differences in belief updating thus manifest directly as differences in the corresponding CPTs, and clustering individuals reduces to clustering their estimated CPTs. We next describe the data sampling process in more detail.

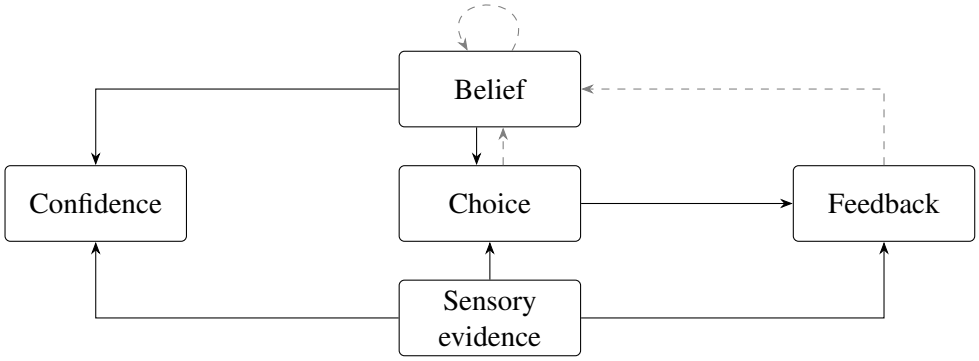


Figure 1. In this graph of the DBN of the experiment, rectangles represent variables and arrows imply causal influences divided into solid lines for influences within the same trial (t) and dashed lines for influences on variables in the next trial ($t + 1$).

3.2 Individual Differences and Groups in the Data

Sensory evidence is sampled randomly. Feedback is deterministic (*correct* if choice and sensory evidence are identical, otherwise *incorrect*). The CPTs for the other variables represent probabilistic associations and can differ between participants.

It is very clear that people differ from each other in the population, among others when it comes to cognitive processes such as belief updating. However, despite individual differences, we assume it is possible to cluster people into different groups. Although the actual groups remain unknown, we have attempted to make a few reasonably plausible assumptions. As we are mainly testing suitability in this study, it is not essential that our defined groups correspond exactly to the actual ones. They should, however, appear plausible to minimise the gap between theory and application.

We define four groups within *belief* updating. (1) Some individuals make *choice-based* updates, i.e., they update beliefs more after correct choices, especially for smaller updates, but rarely adjust after incorrect choices. (2) Others have the same tendency but seem more *insecure* and show more random updating behaviour, i.e., do not seem to have a strong model and switch more between different options. (3) *Sceptical* people have the tendency to update more slowly in a positive direction (e.g., towards greater cooperativeness) whereas (4) some people *persist* in their belief and hardly update it at all despite any evidence or feedback.

For *choice* we assume that there are six different groups. (1) People who are prior focussed (i.e., the probabilities in their CPTs are much higher for combinations where belief and choice are similar) whereas (2) for others it's the opposite, i.e., they are evidence sensitive. Since we assume that there might not only be those two rather extreme groups — which would probably be much easier for clustering — we consider three groups in between: (3) rather prior focussed and (4) rather evidence sensitive, both with similar tendencies as (1) and (2), respectively, but with lower probabilities, and (5) an equal group which has a 50:50 chance for choosing an object matching either the belief or the sensory evidence if the information is inconsistent. (6) The last group differs from this pattern and includes a negativity bias in updating. If prior belief and sensory evidence are not consistent, always the lower value is chosen (e.g., if belief is a *high-value* object and evidence is a *low-value* object, the latter is chosen). People in this group have some updating probabilities equivalent to (1), others to (2) — depending on the direction of updating.

Confidence is only divided in three different groups: (1) Those who are generally rather certain, (2) rather uncertain or (3) specifically certain with their choice if they have a positive prior (i.e., assume that the initiator chooses a *high-value* object).

For each participant, first a group is sampled for each variable. Each group has their own CPT and for each participant, the CPTs of the respective groups are used to sample the data. We incorporate different prior probabilities for the groups, resulting in a structured yet probabilistic data generation process. Thus, the exact data for the individuals will differ from each other despite identical trials (i.e., sensory evidence) due to the probabilistic associations. We aim to make the simulation plausible by incorporating realistic challenges such as continuous transitions between groups, varying numbers of groups, and rare groups that are difficult to detect in limited datasets. This is intended to reduce the gap between theory and practice and to improve the applicability of our findings to real-world data. Although we are aware that there might be additional groups and that the simulated groups and probabilities may differ from reality and cannot capture all aspects of real-world data, we consider them sufficient for evaluating different clustering approaches under realistic circumstances. The goal is not to prove the exact groups we implemented in our simulation but rather to examine whether the groups can be reliably identified from data.

The whole sampling and subsequent analysis has been implemented in R (R Core Team, 2026) using the package *tidyverse* (Wickham et al., 2019) and our implementation is available on GitHub: `https://github.com/R-oadrunner/Hierarchical-Clustering-in-BNs`

4. (Hierarchical) Clustering of Individuals

Fitting one model per participant captures individual updating processes best but does not generalise to the population, while averaging over all individuals ignores genuine differences. Clustering provides a compromise: participants with similar probabilistic associations (e.g., how they update their belief) are grouped together, while differences across groups remain.

To cluster individuals, we build on HACP, the hierarchical agglomerative approach for lifted model construction recently proposed by Speller et al. (2025). Originally devised to compress a factor graph by grouping factors with approximately matching distributions, its agglomerative procedure applies to any collection of objects equipped with a distance and hence also to our setting. Two factors $x, y \in \mathbb{R}^n_{>0}$, viewed as vectors of n positive real numbers, are *ε-equivalent* if their entries agree up to a relative factor of $(1 \pm \varepsilon)$, and the *one-dimensional ε-equivalence distance (1DEED)* returns the smallest such ε, i.e., $D_{\text{1DEED}}(x, y) = \max_{k=1,\dots,n} \frac{|x_k - y_k|}{\min\{x_k, y_k\}}$, where x_k denotes the k-th entry of the vector x. In our setting, the vectors x and y are given by the entries of CPTs. Starting from each object (here: CPT) in its own group, HACP repeatedly merges the closest groups (i.e., CPTs) under this distance. Crucially, it is *hyperparameter-free*: rather than fixing a tolerance ε beforehand, it computes the entire ordered sequence of ε values at which merges occur, producing a hierarchy of progressively coarser groupings that is consistent across levels (objects grouped at some ε stay grouped for every larger ε). We transfer this approach from factors to CPTs, applying the agglomerative merging to distances between participants' CPTs and as the tolerance ε grows, participants with similar probabilistic behaviour are merged first — thus, cutting the resulting hierarchy at a chosen ε yields a concrete partition into groups without having to commit to the number of groups in advance. A major advantage of HACP is that it can also be applied to compute groups of participants in settings where graph structures differ between participants (HACP is naturally designed for such scenarios due to its origin in lifted model construction). Two questions structure the remainder of this section: which distance between CPTs captures the similarity we are interested in, and how to choose the cut-off ε when the true number of groups is unknown.

4.1 Comparison of Distance Metrics

To assess the suitability for our application, we consider different metrics. Specifically, we look at 1DEED (because this has been used in the original HACP paper) as well as the absolute sum of differences $= \sum_{i=1}^{L} |x_i - y_i|$ and the Root Mean Squared Error (RMSE) $= \sqrt{\frac{1}{L}\sum_{i=1}^{L}(x_i - y_i)^2}$ as distances between two vectors x and y with a certain length L. We take three probability vectors (representing three CPTs), x, y, and z as an example:

$$x = \begin{bmatrix} 0.80 \\ 0.19 \\ 0.01 \end{bmatrix} \quad y = \begin{bmatrix} 0.78 \\ 0.17 \\ 0.05 \end{bmatrix} \quad z = \begin{bmatrix} 0.40 \\ 0.59 \\ 0.01 \end{bmatrix}$$

Intuitively, we can see that x and y seem to be quite similar if we consider all probabilities and compare them pairwise. In contrast, z differs largely from both of them in two out of three probabilities. If we calculate the distances for the three metrics mentioned before, we get the following

three distance matrices (whose rows and columns are indexed by x, y, and z):

$$\mathbf{D}_{\text{1DEED}} = \begin{bmatrix} 0.00 & 4.00 & 2.11 \\ 4.00 & 0.00 & 4.00 \\ 2.11 & 4.00 & 0.00 \end{bmatrix} \quad \mathbf{D}_{\text{RMSE}} = \begin{bmatrix} 0.00 & 0.03 & 0.38 \\ 0.03 & 0.00 & 0.33 \\ 0.38 & 0.33 & 0.00 \end{bmatrix} \quad \mathbf{D}_{\text{Sum}} = \begin{bmatrix} 0.00 & 0.08 & 0.80 \\ 0.08 & 0.00 & 0.84 \\ 0.80 & 0.84 & 0.00 \end{bmatrix}$$

Interestingly, we can see that 1DEED implies that x and z are most similar whereas the distance between x and y is much larger. The latter is due to the large impact of the high relative increase of the last probability from 0.01 to 0.05. Since the relative increase is less important in our case than the overall average similarities considering all single values, the other two metrics are a more natural fit here. The RMSE and the sum of the absolute values seem to work equally fine. We decided to use the RMSE because the mean adjusts for the number of data points, making the values more easly interpretable, and penalises larger deviations more by squaring them.

Despite the existence of other well-known distance metrics, we selected the RMSE for comparing CPTs because the objective is to quantify *element-wise* numerical deviations rather than differences in information content or probability mass transport. As described before, the RMSE assigns the same importance to equal absolute deviations regardless of where they occur in the probability range; for example, deviations from 0.00 to 0.05, 0.50 to 0.55, and 0.95 to 1.00 all contribute equally to the error. This property is desirable in the present study because every deviation between corresponding CPTs entries should be weighted solely according to its magnitude, independent of the underlying probability level. In contrast, the *Kullback-Leibler (KL)* divergence (Kullback & Leibler, 1951) quantifies differences in terms of relative information rather than absolute numerical deviations, causing changes involving small probabilities to receive disproportionately larger penalties (e.g., a change from 0.00 to 0.05 is penalised much more strongly than a change from 0.50 to 0.55, while assigning probability zero to an event with non-zero probability results in an infinite divergence). Furthermore, the KL divergence is inherently asymmetric, making it unsuitable for the present application because the objective is to quantify the difference between two individuals, where the distance from person A to person B should be identical to the distance from person B to person A. The *Jeffreys* divergence (Jeffreys, 1946) and the *Jensen-Shannon* divergence (Lin, 1991) overcome this asymmetry. However, both continue to weight probability differences according to their information content rather than treating all equal absolute deviations equally.

Overall, RMSE provides the most appropriate, symmetric, and directly interpretable measure for comparing CPTs when the objective is to quantify only the magnitude of element-wise numerical deviations, irrespective of the probability level at which they occur.

4.2 Application of (Hierarchical) Clustering to an Example Data Set

In this section, we present different approaches and discuss their feasibility, advantages, and disadvantages demonstrated on a simulated example data set with $N = 120$ participants and 420 trials per participant. Variations in these parameters are discussed in Sec. 4.3.

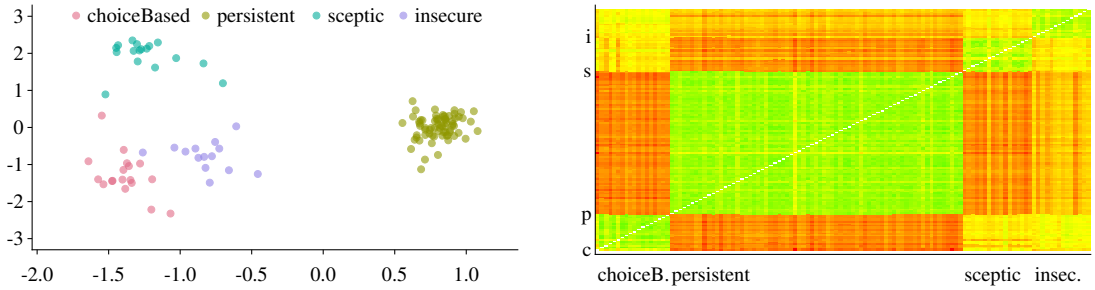
Assuming we know the true number of groups that should be found, we can simply apply the hierarchical clustering explained at the beginning of Sec. 4, which leads, in our data set, to groups that almost exactly match the actual groups from the population for all three variables (*belief*, *choice*, *confidence*). Calculating the average of all individual CPTs within each group (i.e., the mean of

every single probability over all participants belonging to that group) then provides a good estimate of the actual CPTs of a group on the population level.

However, we usually do not know the number of groups and thus, a crucial point in hierarchical clustering is to decide when to stop. Otherwise, all participants would end up in one cluster. One possibility is to set a fixed epsilon as cut-off and whenever epsilon would increase over that threshold, the process stops. But choosing this threshold is challenging if the underlying data structure is unknown and the value might depend on the exact variables, probabilities, and number of groups, participants, and trials. We will critically reflect on different approaches in the following sections.

4.2.1 Visualisation of Distances Between Individuals

As an intuitive approach to get a better overview, it is helpful to plot all individuals with their RMSE distances to each other. We applied multidimensional scaling to the distances in the distance matrix using the function cmdscale in R (R Core Team, 2026). For our example data set, the corresponding plot for *belief* is shown in Fig. 2(a). The true groups of the individuals are indicated by different colours. Visual inspection provides an initial indication of potential groupings within the data but we can also see that it is not always easy to draw a clear line. Especially two groups (choice-based and insecure) are close to each other, which makes sense considering how we set the CPTs in Sec. 3.2.



(a) (b)

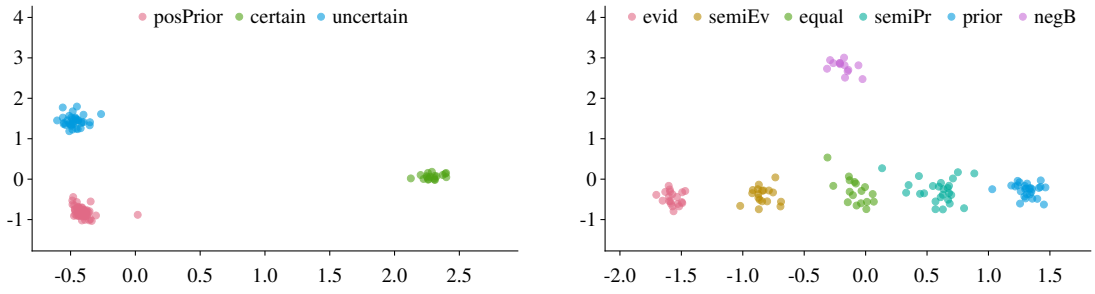
Figure 2. Distances between individuals in a data set are represented (a) as distances in a two-dimensional space and (b) in a heatmap with colours ranging from green (very close) to orange (larger distance); individuals are sorted according to one of the four real groups to which they belong in order to facilitate interpretation.

In Fig. 2(b), the distances are plotted in a heatmap. Individuals are grouped along the axes according to their actual groups. There are also differences within groups, but distances between groups are clearly visible. However, without ordering individuals on the axes, heatmaps can become confusing and difficult to interpret, especially in larger samples.

The variables *belief*, *choice*, and *confidence* are differently easy to cluster. The easiest is *confidence* because there are only three rather easily distinguishable groups. *Belief* and *choice* are more difficult — the latter mainly because of the number of groups (six) and the continuous order of groups whereas *belief* has two groups that have a higher similarity as described in Sec. 3.2.

Although we mainly focus on the variable *belief* in most parts of this paper, the other variables will also be considered for the comparison of the different methods when varying the number of

participants and trials in Sec. 4.3. In Fig. 3, we therefore visualise the distances of the groups of the other two variables — *confidence* and *choice* — described in Sec. 3.2 for the same data set. It becomes evident that the three groups of *confidence* are rather easily distinguishable whereas the structure of *choice* seems more complex. Five of the six groups of *choice* vary from *evidence sensitive* over *semi evidence sensitive* and *equally evidence and prior based* to *semi prior focussed* and *prior focussed*, which visually demonstrates the idea we considered during the sampling procedure and definitions of the CPTs. Only *negative bias* is separate from this continuous scale. Due to the proximity of groups, it can happen that some individuals are hard to cluster, especially if the within-group variance increases with fewer trials.



(a) (b)

Figure 3. Distances between individuals in one data set are visualised in a two-dimensional space for (a) *confidence* and (b) *choice* with colours indicating true groups. Groups in (a) are *positive prior*, *certain*, and *uncertain* and in (b) are *evidence sensitive*, *semi evidence sensitive*, *equally evidence and prior based*, *semi prior focussed*, *prior focussed*, and *negative bias*.

4.2.2 Elbow Plot to Support Finding the Best Epsilon

Figure 4 visualises the increase in ε for step 1 to $N-1$ of the clustering process (where N is the number of individuals to be clustered) for the variable belief. The greater the increase, the larger the distance between the two nearest individuals or groups that are clustered in this step. Increases are mostly small during large parts of the process but towards the end, ε increases largely. After the last step ($N-1$), all individuals are in the same cluster. As known from other applications, this elbow plot can be used to determine the optimal number of groups.

In ambiguous plots, one can also consider the degree of caution they would like to take for deciding when to stop the clustering. Stopping too early might lead to homogeneous but too many groups and thereby inefficient data aggregation whereas stopping too late might end up with heterogeneous groups and might cluster actually distinct groups together. To support the decision, distances of the final or multiple potentially good clustering steps can be plotted similar to Fig. 2 to give hints about potential clustering options and provide information about the constellation of the groups, e.g., which groups are closer than others.

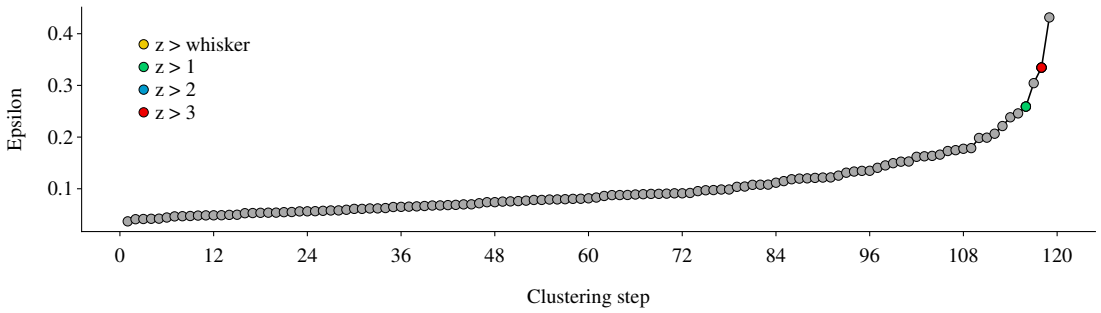


Figure 4. Increase in ε (y-axis) is plotted in dependence of the clustering step (x-axis). Coloured dots indicate potential cut-off steps depending on the stopping criterion. As steps may satisfy several cut-off values, only the strictest one is colour-coded. This is the reason why there are no corresponding coloured dots for $z >$ whisker and $z > 2$ in this specific case.

4.2.3 Evaluation of Different Epsilon Cut-Offs

Choosing a fixed value for ε turned out to be difficult because ε differs depending on the chosen variable as well as the number of participants and trials. We tried different cut-off values such as an absolute value for ε, a multiple of the mean increases until the current step, the previous maximum increase, several times the sum of the increases in the previous s steps, the fit of a point of the ε curve within a linear model from previous to following steps, quantiles, or outliers identified by z-scores, as well as variations and different combinations of these approaches. They all turned out to be overfitted for specific data sets or parameter settings. But since the relationship between certain data set specific aspects or parameter settings and the optimal cut-off value was not completely reliable and itself depended on the complexity of the data set (e.g., number of groups, variance in the data, number of possible values of a variable), attempts to fix it ended in reduced generalisability for different data sets. We also tested to choose a low threshold and define the earliest step to stop (i.e., the maximum number of groups) but in large data sets with many trials this almost always resulted in the maximum number representing the final grouping. Additionally, this maximum needs to be chosen beforehand which we do not consider to be an optimal strategy since we want to reduce the assumptions that need to be made.

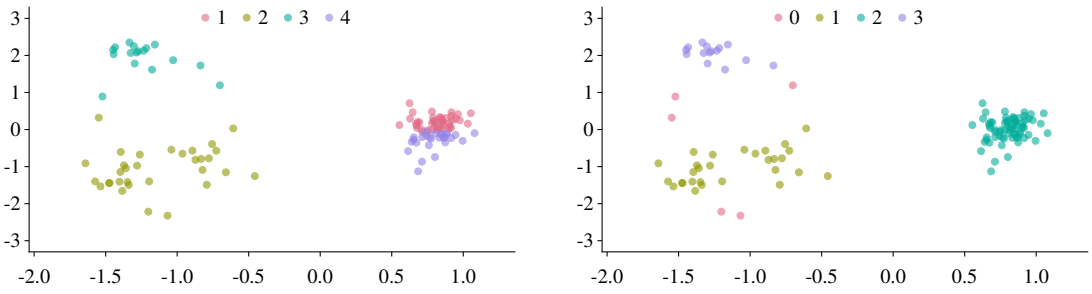
To reduce the impact of the factors mentioned before, we only used the last 25 steps to identify the cut-off. We chose 25 because we can keep this number fixed and use the same number of steps for all data sets — independent of the number of participants — even for small data sets, e.g., with $N = 30$ participants. We calculated the increase of ε from one step to the next one for these 25 steps only and scaled this data. Then, we defined differently strict outliers, mainly using z-scores (i.e., standard deviations (SDs) of the scaled distribution) $z > 1$, $z > 2$, and $z > 3$, as well as typical outliers used for boxplots which are values exceeding the upper fence, which defines the maximum extent of the upper *whisker*, i.e., the 75% quantile $(Q_3) + 1.5 \cdot$ Interquartile Range (IQR).

The outliers identified by the z-values or whisker of the scaled data are marked in Fig. 4 by the differently coloured dots. In this plot, not all of the possible cut-off values are present. Please note that $z > 3$ also implies $z > 2$ and $z > 1$ and thus, only the strictest cut-off (here $z > 3$)

is highlighted. Depending on the variable we looked at and the concrete data set, the best cut-off varied but these different values often gave at least some good indication about the range of the possible number of groups (discussed in more detail in Sec. 4.3).

4.2.4 K-Means and DBSCAN

For comparison, we tested k-means using the function kmeans in R (R Core Team, 2026) with scaled RMSE distances. We set k (i.e., the number of groups) to the real number of groups in the corresponding data set. The clustering by k-means for our illustration data set visualised in Fig. 2(a) is shown in Fig. 5(a). Comparing both figures reveals obvious differences. However, k-means is sensitive to initialisation and produces different cluster structures and varying performance across repeated runs on the same data set.



(a) (b)

Figure 5. Distances between the individuals of one data set are shown in a two-dimensional space with colours indicating the clusters found by (a) k-means and (b) DBSCAN.

We also used DBSCAN with the same scaled RMSE distances and the function dbscan (Hahsler & Piekenbrock, 2025) with results shown in Fig. 5(b). We set the minimum number of participants per group to 3 and set $\varepsilon = 0.5$ after testing for several data sets. However, the quality varied from having only one group to far too many groups and sometimes a lot of outliers.

4.3 Impact of Different Numbers of Participants and Trials on Clustering

To test the previously explained approaches on different data sets, we constructed sequences of parameters using logarithmically spaced values to cover a wide range of magnitudes in a balanced way since increases in larger numbers are considered to have a lower impact than increases in lower numbers. For the number of participants we tested $N \in \{30, 60, 120, 250, 500\}$ with different numbers of trials, specifically $n_{\text{trials}} \in \{50, 80, 120, 180, 280, 420, 650, 1000\}$. All values of N were combined with all values of n_{trials} and repeated 50 times each resulting in a total of 2000 data sets to ensure a variety of data sets.

We compare different approaches: (1) – (5) for hierarchical clustering using different stopping criteria for the clustering process as well as (6) k-means and (7) DBSCAN. (1) has (just like k-means) the unfair advantage of "knowing" the real number of groups existent in the current data set. (2) – (5) use the scaled data of the last 25 steps as explained in Sec. 4.2.3, specifically (2) the

maximal boxplot whisker ($Q_3 + 1.5 \cdot \text{IQR}$), (3) $z > 1$, (4) $z > 2$, and (5) $z > 3$. All of them were applied to all 2000 data sets. We did not include the visual elbow method to determine the cut-off as it would have been (a) too time-consuming and (b) subjective.

4.3.1 Fit of Found Clusters Compared to True Groups

To quantify how homogeneous a cluster is considering the true groups of its individuals, we take the largest relative frequency of any group in a cluster, i.e., the maximum is 1. To calculate the average fit across all identified clusters for one variable, the average is taken. In concrete terms, this means $\text{fit}_{\text{raw}} = \frac{1}{|C|} \sum_{c \in C} \max_{g \in G} p(g \mid c)$ with G being all true groups, $|C|$ the number of clusters identified from data, and $p(g \mid c)$ the relative frequency of the number of individuals belonging to group $g \in G$ being assigned to cluster $c \in C$. For example, for 4 perfectly identified groups, the fit would be $\frac{1+1+1+1}{4} = 1$, i.e., a perfect fit. In the worst case, all groups are evenly distributed, i.e., the fit is $\frac{1}{|G|}$, where $|G|$ is the number of true groups. To make the fit independent of $|G|$, we normalise the values to the range $[0, 1]$ by

$$\text{fit}_{\text{norm}} = \frac{\text{fit}_{\text{raw}} - \frac{1}{|G|}}{1 - \frac{1}{|G|}}$$

with 1 still being a perfect fit and 0 being the worst case of maximum equal distribution.

In Fig. 6, the normalised mean fit (with error bars for the SD resulting from the different data sets) is shown for the hierarchical clustering stopping at the actual number of true groups.

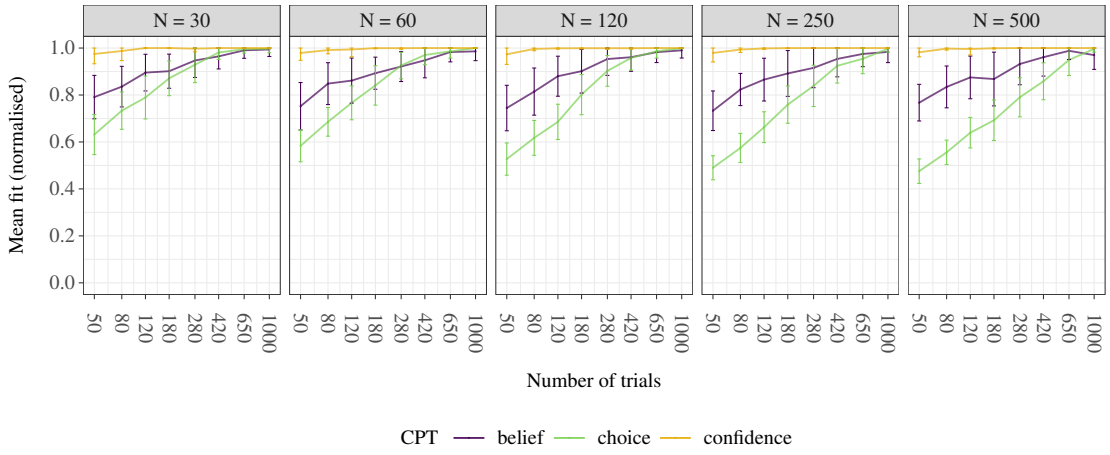


Figure 6. The normalised mean fit is shown for the hierarchical clustering stopping at the actual number of groups split for N (grey boxes on the top) and n_{trials} (x-axis). Error bars represent the variation between different data sets, given by the SD. The different colours belong to the three different variables *belief*, *choice*, and *confidence*, for which the groups have been estimated.

The average fit increases and the SD decreases with increasing number of trials for all variables, although the increase is barely visible for confidence because the fit is already almost perfect for the lowest number of trials that we have tested (50). With 1000 trials, the fit is on average almost at 1

(i.e., perfect) for all variables. The number of participants has little impact. Considering the mean fit for the different number of participants with 50 trials, an increase in the number of participants might on average even have a small negative effect in data sets with fewer trials.

An increase in the number of trials leads to more easily distinguishable groups and a more easily visible cut-off value in the elbow plot. To visualise the effect of the number of trials, we selected two additional data sets. Similar to the data set we used for illustration in Sec. 4.2, we held the number of participants constant at $N = 120$ but chose two different numbers of trials. In Fig. 7, the distances of the two data sets are plotted in a two-dimensional space for 50 trials in Fig. 7(a) and 1000 trials in Fig. 7(c). The existing true groups are identical but the variance within the groups is much higher in the data set with fewer trials which makes clustering much more difficult. The corresponding elbow plots when applying hierarchical clustering to both data sets visualised in Figs. 7(a) and 7(c) are shown in Figs. 7(b) and 7(d), respectively.

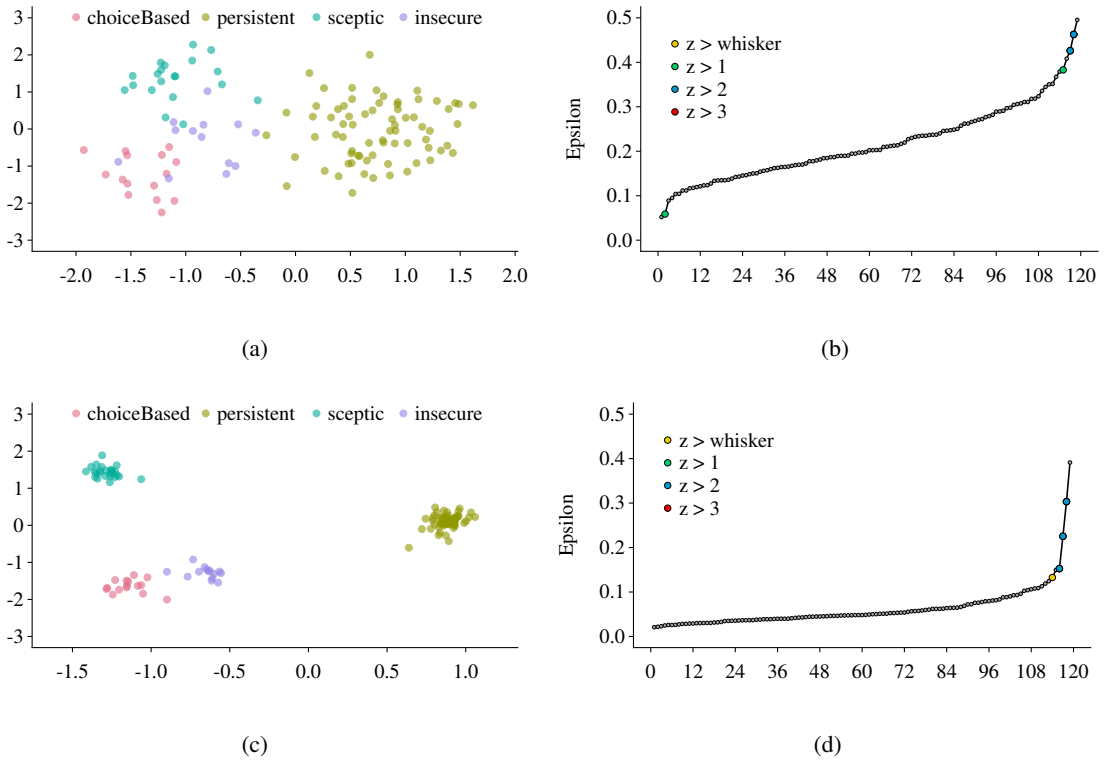


Figure 7. Distances between individuals in two different data sets, both with $N = 120$ participants, are shown in a two-dimensional space on the left for (a) 50 and (c) 1000 trials per participant, respectively. The elbow plots on the right visualise the increase in ε when applying hierarchical clustering on these two data sets, for (b) 50 and for (d) 1000 trials per participant with different outlier-based cut-off values highlighted as coloured dots. Since some steps may satisfy several cut-off values, only the strictest one is colour-coded. Therefore, not every possible cut-off has a corresponding coloured dot.

In the sample with more trials, the elbow is much better visible, the increases seem quite stable, and the curve is rather flat until almost the end, i.e., until different groups need to be clustered

together ε does not need to be increased much in order to cluster individuals (of the same group). For fewer trials, the curve is a bit less flat and there are more visible bumps in the increases. Additionally, the last increases are not as easily distinguishable from the rest of the curve as with more trials.

Overall, the impact of the number of trials is lowest for *confidence* as the fit is already good, probably due to the smaller number of groups and the good distinguishability between them. For variables with more similarities between different groups, clustering becomes more difficult.

Next, we consider the fit for the groups determined by the different methods. We excluded outliers from DBSCAN (on average 1.28 outliers, SD $= 2.20$, median $= 0$, maximum $= 18$) because considering all of them as one group would largely reduce the overall fit. Since the sample size does not seem to have a large impact, we do not split the results according to the number of participants. Figure 8 shows the average fit including bars for the SD to illustrate variation between different data sets, split by the three variables. For all methods, it worked best for *confidence* with DBSCAN being slightly worse with a smaller number of trials, k-means having a larger SD than most other methods, and $\varepsilon_{z>3}$ performing worst of all methods with SD increasing largely for a decreasing number of trials. Differences between the methods become more clearly visible for the other two variables, especially for *choice*. $\varepsilon_{z>3}$ again has the lowest fit and is the only method not improving much with more trials. The best average performance can be seen for ε stopping at the real number of groups, $\varepsilon_{Q_3+1.5\cdot IQR}$ (having the boxplot whisker as cut-off), and k-means, of which $\varepsilon_{Q_3+1.5\cdot IQR}$ is the only one not having the advantage of including the knowledge about the real number of groups. The fit gets worse with later stopping of the hierarchical clustering which might be explained by an underestimation of the number of groups.

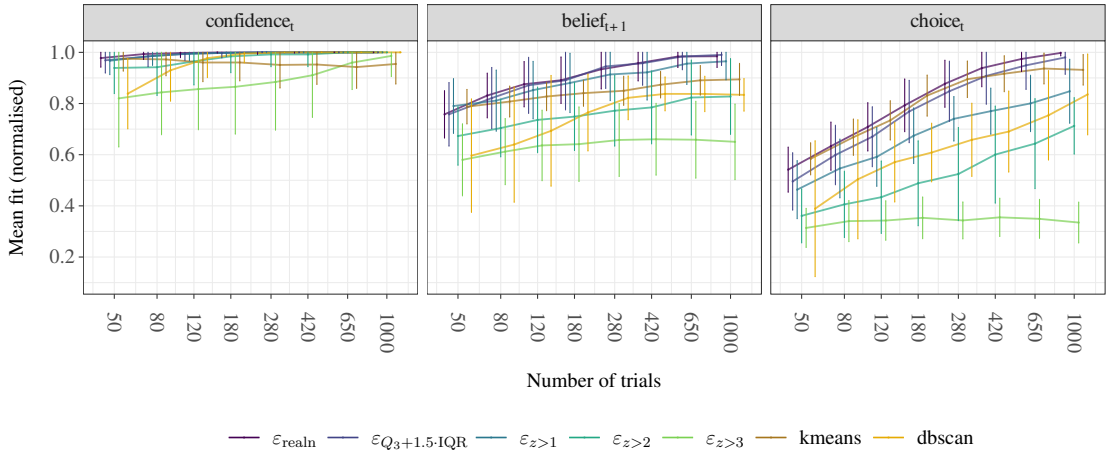


Figure 8. The normalised mean fit (y-axis) is shown in dependence of the number of trials (x-axis) for different methods (colours) and SD, i.e., variation between data sets (error bars) for the three variables *confidence*, *belief*, and *choice* (boxes).

4.3.2 Number of Found Groups

Another evaluation criterion is the number of clusters being identified. We calculated the difference between the number of identified clusters and the actual number of groups in the data, $n_{\text{clusters}}(\text{data}) - n_{\text{groups}}(\text{real})$, where the optimal value is 0. Values above 0 indicate an overestimation of the number of groups (too many clusters were identified). A negative value, on the other hand, means that the algorithm has identified too few groups, thus reduced the data too strongly.

Again, we removed outliers from DBSCAN clustering and did not count them as a separate group. Results for k-means and the hierarchical clustering stopping at the correct number of groups are not included since these two methods have the inherent property that the deviation is zero. As Fig. 9 demonstrates, the estimation differs between the three variables with the tendency to underestimate the number of groups with an increasing number of true groups and complexity in the data (e.g., *choice* compared to *confidence*). The whisker cut-off tends to end up with the highest number of groups (with a large variation between the data sets), leading on average to an overestimation of the number of groups in *confidence* but a better estimation in *choice* compared to the other methods. The number of trials does not seem to be as important as for the fit regarding the mean estimation but slightly reduces the SD for most methods and variables.

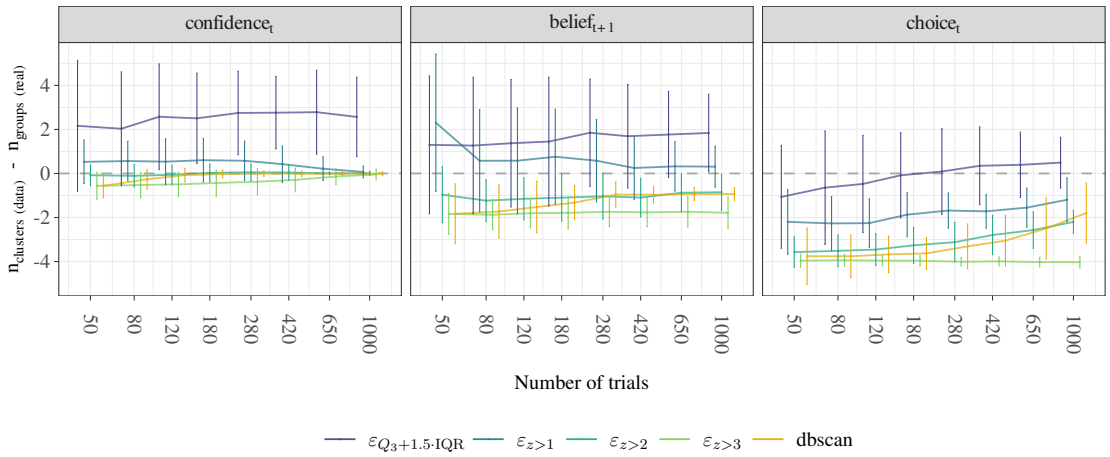


Figure 9. Differences between the number of clusters and real groups (y-axis) for different number of trials (x-axis) and methods (colours) are shown for each variable (boxes) with SD (error bars) between data sets.

5. Discussion

We have presented an effective solution to the previously unaddressed problem of clustering participants based on similar cognitive processes where a separate model is fitted per individual and groups of individuals with similarities in belief updating (or other processes) are sought. Our simulation study demonstrates that the clustering approach reliably recovers group structures in cognitive processes which allows for inter-individual differences whilst ensuring generalisability and minimising overfitting to individuals. Hierarchical clustering avoids committing to the number of groups in

advance, as in, e.g., k-means, or certain parameter settings, as in e.g., DBSCAN but rather provides a fully inspectable merge sequence which nevertheless allows users to make their own decisions regarding the balance between the strength of aggregation and the homogeneity of the clusters.

The inclusion of a social-cognitive perspective does not change the clustering procedure itself, but serves to establish a link between the algorithmic approach and real-world phenomena. In particular, modelling belief updating as a cognitive process provides an example through which the general applicability of the approach can be examined beyond the abstract algorithmic level. This perspective therefore helps to bridge the gap between the identification of patterns in dynamic Bayesian networks and their implementation and interpretation in terms of processes that may occur in real-world settings. Importantly, the framework is not limited to belief updating and could be extended to other cognitive processes, provided that the underlying assumptions are met: individuals may differ in their cognitive or behavioural patterns, while at the same time exhibiting sufficient similarities for meaningful clusters to be identified.

Some limitations remain and include dependence of clustering performance on experimental design, with low trial counts leading to unstable results in complex settings. The study is restricted to a single experimental setup and causal structure, making it difficult to disentangle effects of group numbers and group proximities. Although we intentionally varied the number of groups for the variables, the simultaneous variation in group proximity unfortunately prevents a systematic and unambiguous analysis (i.e., one independent of the proximity of the groups) of the effect of the number of groups. Our current simulated data sets also omit unobserved variables that may affect underlying data structures and we assume that there are separable groups. Using a single parameter setting and removing outliers may introduce bias in DBSCAN results and evaluating k-means only with a known number of clusters reduces realism. Future research should improve generalisability by incorporating unobservable variables, different causal structures and a wider range of ε-selection methods including more robust approaches for choosing elbow plot cut-off points, such as automated or visually guided criteria.

Our next step is to apply the knowledge we have gained to the real study. We have now laid the foundations for bridging the gap between theory and practice and in doing so, we have identified relevant points that should be taken into account. We will use this knowledge, for example, to design our experiment accordingly (e.g., with regard to the number of trials), to define the distances in line with our requirements, and to be able to cluster individuals based on their similarity in terms of belief updating mechanisms using the clustering approach that we have tested.

6. Conclusion

Hierarchical clustering of individuals with similar cognitive processes generally performs well even with realistic, smaller datasets, though its robustness depends on the data structure and the number of trials. The ε-based method is transparent, requires no prior specification of cluster numbers and yields consistent results across runs. Updating distances during clustering improves interpretability by clarifying relationships between groups, and the approach remains computationally efficient even for large datasets. However, results depend strongly on the choice of the distance metric, which must match the application goal. Determining the optimal number of clusters is still difficult, particularly

when clusters are overlapping. This introduces a trade-off between more homogeneous clusters and more compact data aggregation, making theory-guided support of the clustering process and interpretation advisable when it is applied to actual data to model cognitive processes.

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